Spinors & The Dirac Equation

Peskin & Schroeder : An Introduction to Quantum Field Theory § 3.1~3.2

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1. Lorentz Invariance in Wave equations

1-1. Scalar Field

What do we mean by “relativistically invariant”?

$$x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}, \quad y \equiv \Lambda^{-1} x$$

$$\varphi'(x) = \varphi(y)$$

The scalar has the same value at corresponding spacetime points.

$$\partial_{\mu} \varphi'(x) = (\Lambda^{-1})^{\nu}_{\mu} (\partial_{\nu} \varphi)(y)$$

The chain rule determines how derivatives transform.

$$g^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$$

1. Lorentz Invariance in Wave equations

1-1. Scalar Field

$$g^{\mu\nu}\partial_\mu\varphi'(x)\partial_\nu\varphi'(x)$$

$$= g^{\mu\nu}(\Lambda^{-1})^\rho_\mu(\Lambda^{-1})^\sigma_\nu(\partial_\rho\varphi)(y)(\partial_\sigma\varphi)(y)$$

$$= g^{\rho\sigma}(\partial_\rho\varphi)(y)(\partial_\sigma\varphi)(y)$$

Lorentz transformations preserve the metric.

$$g^{\mu\nu}(\Lambda^{-1})^\rho_\mu(\Lambda^{-1})^\sigma_\nu = g^{\rho\sigma}$$

The transformed Klein–Gordon equation has the same form.

$$(\partial^2 + m^2)\varphi'(x) = [(\partial^2 + m^2)\varphi](y) = 0$$

1. Lorentz Invariance in Wave equations

1-2. Vector Field

A vector transformation also mixes field components.

$$V'^{i}(x) = R^{i}_{j} V^{j}(R^{-1}x)$$

$$V'^{\mu}(x) = \Lambda^{\mu}_{\nu} V^{\nu}(\Lambda^{-1}x)$$

The matrix acts on the index. The argument changes as for a scalar.

$$(\partial^2 + m^2)V'^{\mu}(x) = \Lambda^{\mu}_{\nu}[(\partial^2 + m^2)V^{\nu}](y)$$

A representation specifies how the components transform.

1. Lorentz Invariance in Wave equations

1-2. Vector Field: rotations

Different matrices can obey the same angular-momentum algebra.

$$[J^i, J^j] = i\varepsilon^{ijk} J^k$$

$$R = exp(-i\theta^i J^i)$$

$$\mathbf{J} = \mathbf{x} \times (-i\,\boldsymbol{\nabla})$$

$$J^i = \frac{\sigma^i}{2}, \quad U = exp(-\frac{i}{2}\boldsymbol{\theta} \cdot \boldsymbol{\sigma})$$

The Pauli matrices give a two-component representation.

1. Lorentz Invariance in Wave equations

1-2. Vector Field: Lorentz generators

$$J^{\mu\nu} = i(x^{\mu}\partial^{\nu} - x^{\nu}\partial^{\mu})$$

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma}$$

$$-g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho})$$

The four-vector representation uses 4 × 4 matrices.

$$(\mathcal{J}^{\mu\nu})^{\alpha}_{\beta} = i(g^{\mu\alpha}\delta^{\nu}_{\beta} - g^{\nu\alpha}\delta^{\mu}_{\beta})$$

$$\Lambda = exp(-\frac{i}{2}\omega_{\mu\nu}\mathcal{J}^{\mu\nu}), \quad \omega_{\mu\nu} = -\omega_{\nu\mu}$$

1. Lorentz Invariance in Wave equations

1-2. Vector Field: rotation and boost

To first order in the transformation parameters:

$$V'^{\alpha} = (\delta^{\alpha}_{\beta} - \frac{i}{2}\omega_{\mu\nu}(\mathcal{J}^{\mu\nu})^{\alpha}_{\beta})V^{\beta}$$

$$\omega_{12} = \theta: \quad V' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\theta & 0 \\ 0 & \theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V$$

$$\omega_{01} = \beta: \quad V' = \begin{pmatrix} 1 & \beta & 0 & 0 \\ \beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V$$

2. Spinor representation

2-1. Dirac Matrices

We seek matrices that generate another Lorentz representation.

$$\{\gamma^{\mu}, \gamma^{\nu}\} = 2g^{\mu\nu}1_4$$

$$\gamma^0 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

This is the Weyl (chiral) basis. Each block is 2 × 2.

$$(\gamma^0)^2 = 1_4, \quad (\gamma^i)^2 = -1_4, \quad \{\gamma^0, \gamma^i\} = 0$$

2. Spinor representation

2-1. Dirac Matrices: spinor generators

$$S^{\mu\nu} \equiv \frac{i}{4}[\gamma^{\mu}, \gamma^{\nu}]$$

The Clifford algebra reduces products of three gamma matrices.

$$[\gamma^{\mu}\gamma^{\nu}, \gamma^{\rho}] = \gamma^{\mu}\{\gamma^{\nu}, \gamma^{\rho}\} - \{\gamma^{\mu}, \gamma^{\rho}\}\gamma^{\nu}$$

$$= 2(g^{\nu\rho}\gamma^{\mu} - g^{\mu\rho}\gamma^{\nu})$$

$$[S^{\mu\nu}, \gamma^{\rho}] = i(g^{\nu\rho}\gamma^{\mu} - g^{\mu\rho}\gamma^{\nu})$$

2. Spinor representation

2-1. Dirac Matrices: the Lorentz algebra

Use the commutator from the previous slide.

$$[S^{\mu\nu}, S^{\rho\sigma}] = \frac{i}{4}[S^{\mu\nu}, [\gamma^{\rho}, \gamma^{\sigma}]]$$

$$= \frac{i}{4}([[S^{\mu\nu}, \gamma^{\rho}], \gamma^{\sigma}] + [\gamma^{\rho}, [S^{\mu\nu}, \gamma^{\sigma}]])$$

$$= i(g^{\nu\rho}S^{\mu\sigma} - g^{\mu\rho}S^{\nu\sigma}$$

$$-g^{\nu\sigma}S^{\mu\rho} + g^{\mu\sigma}S^{\nu\rho})$$

The same algebra gives a new representation of Lorentz transformations.

2. Spinor representation

2-1. Dirac Matrices: rotations and boosts

$$S^{ij} = \frac{1}{2}\varepsilon^{ijk}\begin{pmatrix}\sigma^k & 0\\ 0 & \sigma^k\end{pmatrix}$$

$$S^{0i} = -\frac{i}{2}\begin{pmatrix}\sigma^i & 0\\ 0 & -\sigma^i\end{pmatrix}$$

Rotations act the same way on the two blocks.

Boosts act with opposite signs.

$$(S^{ij})^\dagger = S^{ij}, \quad (S^{0i})^\dagger = -S^{0i}$$

2. Spinor representation

2-2. Lorentz Transformations of Spinors

$$D(\Lambda) \equiv \Lambda_{\frac{1}{2}} = exp(-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu})$$

$$\psi'(x) = D(\Lambda)\psi(\Lambda^{-1}x)$$

D acts on spinor components. Λ acts on spacetime indices.

$$D(\Lambda) = 1 - \frac{i}{2}\omega_{\mu\nu}S^{\mu\nu} + O(\omega^2)$$

The gamma matrices connect these two representations.

2. Spinor representation

2-2. Lorentz Transformations of Spinors

First consider an infinitesimal transformation.

$$D^{-1}\gamma^{\mu}D = \gamma^{\mu} + \frac{i}{2}\omega_{\rho\sigma}[S^{\rho\sigma}, \gamma^{\mu}] + O(\omega^2)$$

$$= \gamma^{\mu} - \frac{1}{2}\omega_{\rho\sigma}(g^{\sigma\mu}\gamma^{\rho} - g^{\rho\mu}\gamma^{\sigma}) + O(\omega^2)$$

$$= \gamma^{\mu} + \omega^{\mu}_{\nu}\gamma^{\nu} + O(\omega^2)$$

$$= \Lambda^{\mu}_{\nu}\gamma^{\nu} + O(\omega^2)$$

Antisymmetry of ω combines the two terms.

2. Spinor representation

2-2. Lorentz Transformations of Spinors

Composition extends the identity to finite transformations.

$$D_a^{-1}\gamma^\mu D_a = (\Lambda_a)^\mu_\nu \gamma^\nu$$

$$(D_2 D_1)^{-1}\gamma^\mu (D_2 D_1)$$

$$= (\Lambda_2)^\mu_\nu (\Lambda_1)^\nu_\rho \gamma^\rho$$

$$D(\Lambda)^{-1}\gamma^\mu D(\Lambda) = \Lambda^\mu_\nu \gamma^\nu$$

This identity makes the Dirac equation Lorentz covariant.

3. The Dirac Equation

3-1. Lorentz-invariant Dirac Lagrangian

$$(i\gamma^{\mu}\partial_{\mu} - m)\psi = 0$$

Apply the transformed derivative to ψ′(x) = Dψ(y).

$$(i\gamma^{\mu}\partial_{\mu} - m)\psi'(x)$$

$$= D[i(D^{-1}\gamma^{\mu}D)(\Lambda^{-1})^{\nu}_{\mu}\partial_{\nu} - m]\psi(y)$$

$$= D[i\Lambda^{\mu}_{\rho}\gamma^{\rho}(\Lambda^{-1})^{\nu}_{\mu}\partial_{\nu} - m]\psi(y)$$

$$= D(i\gamma^{\nu}\partial_{\nu} - m)\psi(y) = 0$$

3. The Dirac Equation

3-1. The Klein–Gordon equation

Every component of a free Dirac field satisfies the KG equation.

$$0 = (-i\gamma^{\mu}\partial_{\mu} - m)(i\gamma^{\nu}\partial_{\nu} - m)\psi$$

$$= (\gamma^{\mu}\gamma^{\nu}\partial_{\mu}\partial_{\nu} + m^2)\psi$$

$$= (\frac{1}{2}\{\gamma^{\mu}, \gamma^{\nu}\}\partial_{\mu}\partial_{\nu} + m^2)\psi$$

$$= (\partial^2 + m^2)\psi$$

The symmetric derivatives remove the gamma commutator.

3. The Dirac Equation

3-1. The Dirac adjoint

Boosts are not unitary, so ψ†ψ is not a Lorentz scalar.

$$\overline{\psi} \equiv \psi^\dagger \gamma^0, \quad (\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$$

$$(S^{\mu\nu})^\dagger = \gamma^0 S^{\mu\nu} \gamma^0$$

$$D^\dagger \gamma^0 = \gamma^0 D^{-1}$$

$$\overline{\psi}'(x) = \psi^\dagger(y) D^\dagger \gamma^0 = \overline{\psi}(y) D^{-1}$$

$$\overline{\psi}'(x)\psi'(x) = \overline{\psi}(y)\psi(y)$$

3. The Dirac Equation

3-1. Lorentz-invariant Dirac Lagrangian

$$\mathcal{L}_{\text{Dirac}} = \overline{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi$$

The adjoint and gamma identities make both terms Lorentz scalars.

$$\overline{\psi}' i\gamma^{\mu}\partial_{\mu}\psi' = \overline{\psi} D^{-1} i\gamma^{\mu} D (\Lambda^{-1})^{\nu}_{\mu}\partial_{\nu}\psi$$

$$= \overline{\psi} i\gamma^{\nu}\partial_{\nu}\psi$$

Treat ψ and its adjoint as independent fields in the variation.

$$\delta\overline{\psi}: \quad (i\gamma^{\mu}\partial_{\mu} - m)\psi = 0$$

$$\delta\psi: \quad -i(\partial_{\mu}\overline{\psi})\gamma^{\mu} - m\,\overline{\psi} = 0$$

3. The Dirac Equation

3-2. Weyl Spinors

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$$

Each two-component field transforms independently.

$$\psi_L{}' = (1 - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\sigma} - \frac{1}{2}\boldsymbol{\beta}\cdot\boldsymbol{\sigma})\psi_L$$

$$\psi_R{}' = (1 - \frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\sigma} + \frac{1}{2}\boldsymbol{\beta}\cdot\boldsymbol{\sigma})\psi_R$$

The opposite boost signs distinguish left- and right-handed spinors.

3. The Dirac Equation

3-2. Weyl Spinors: the mass term

In the Weyl basis, the Dirac equation becomes

$$\begin{pmatrix} -m & i(\partial_0 + \boldsymbol{\sigma}\cdot\boldsymbol{\nabla}) \\ i(\partial_0 - \boldsymbol{\sigma}\cdot\boldsymbol{\nabla}) & -m \end{pmatrix}\begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = 0$$

$$i(\partial_0 + \boldsymbol{\sigma}\cdot\boldsymbol{\nabla})\psi_R = m\psi_L$$

$$i(\partial_0 - \boldsymbol{\sigma}\cdot\boldsymbol{\nabla})\psi_L = m\psi_R$$

The mass couples the left and right fields in the equations of motion.

3. The Dirac Equation

3-2. Weyl Spinors: the massless limit

$m = 0$

$i(\partial_0 - \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\psi_L = 0$

$i(\partial_0 + \boldsymbol{\sigma} \cdot \boldsymbol{\nabla})\psi_R = 0$

The two Weyl equations decouple.

A single two-component field can describe a massless fermion.

3. The Dirac Equation

3-2. Weyl Spinors: compact notation

$$\sigma^{\mu} \equiv (1_2, \boldsymbol{\sigma}), \quad \overline{\sigma}^{\mu} \equiv (1_2, -\boldsymbol{\sigma})$$

$$\gamma^{\mu} = \begin{pmatrix} 0 & \sigma^{\mu} \\ \overline{\sigma}^{\mu} & 0 \end{pmatrix}$$

$$i\sigma^{\mu}\partial_{\mu}\psi_R = m\psi_L, \quad i\,\overline{\sigma}^{\mu}\partial_{\mu}\psi_L = m\psi_R$$

For m = 0:

$$i\,\overline{\sigma}^{\mu}\partial_{\mu}\psi_L = 0, \quad i\sigma^{\mu}\partial_{\mu}\psi_R = 0$$

4. Summary

The transformation law determines the type of field.

$$\varphi'(x) = \varphi(y), \quad V'^{\mu}(x) = \Lambda^{\mu}_{\nu} V^{\nu}(y), \quad \psi'(x) = D\psi(y)$$

Gamma matrices connect spinors to Lorentz vectors.

$$D^{-1}\gamma^{\mu}D = \Lambda^{\mu}_{\nu}\gamma^{\nu}$$

The adjoint gives a Lorentz-invariant Dirac Lagrangian.

$$\mathcal{L}_{\text{Dirac}} = \overline{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi$$

The massless Dirac equation splits into two independent Weyl equations.

$$i\,\overline{\sigma}^{\mu}\partial_{\mu}\psi_L = 0, \quad i\sigma^{\mu}\partial_{\mu}\psi_R = 0$$